Managing emerald ash borer outliers

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1 Overview

The emerald ash borer spreads over long distances when people move infested wood, founding outlier colonies far ahead of the main front. Once one is found, what should be done there? Mercader et al. (2011b) took their model of local spread (Mercader et al. (2011a); see the emerald ash borer vignette) and added three treatments in a circle around the origin of an outlier:

- ash reduction: harvesting the large trees, removing their phloem and the larvae in it, once;
- girdling: girdling one large tree a cell each year, which draws egg-laying females to it, and destroying it with its larvae;
- insecticide: protecting the large trees with a systemic insecticide, which kills the larvae laid in them and keeps their phloem from being eaten.

Large trees (27.9 cm in diameter or more) hold half the phloem by default. The authors also asked how the results change if females are drawn to stressed trees.

This implementation is a reproduction of the authors' R code (the article's Supplemental Appendix S1): it follows the code where the code and the text differ, as the deviations below list, and with the code's indexing and the same tolerance for bare cells it gives the code's results cell by cell.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
```

2 The treatments

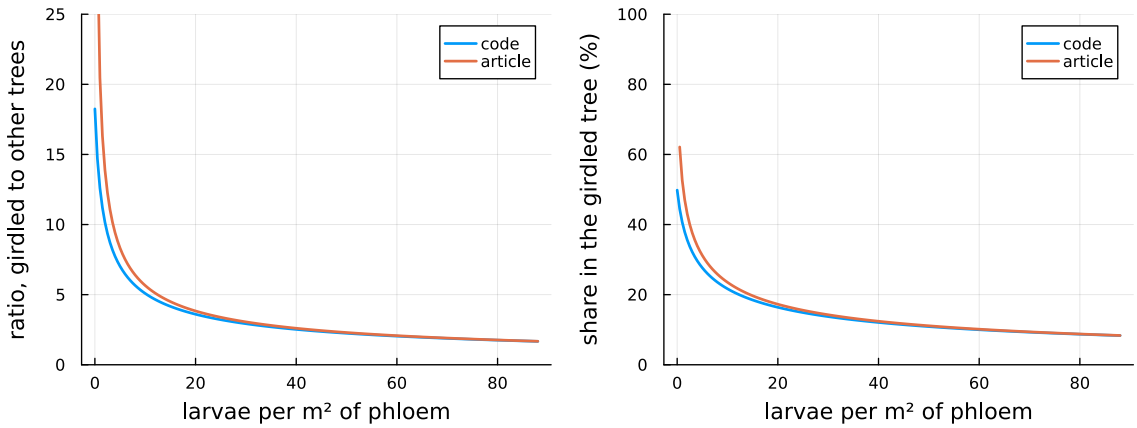
The landscape is 100 × 100 cells of 50 m, each with 200 m² of ash phloem, with 100 adults released at the cell (51, 51). Treated cells lie within a radius of that origin. Without treatment the model is the base model with avoidance, as the code runs it: a dispersal window of 40 cells, no border, and cells with 0.9 adults or fewer dispersing none. Its radius of infestation after 15 years is 1350 m.

```
radii(sol) = [eab_spread(u).radius for u in sol.u]
simulate(; kw...) = solve(problem(:eab_management; kw...); saveat=1)
untreated = simulate()
radii(untreated)'
```

```
1×16 adjoint(::Vector{Float64}) with eltype Float64:
 0.0  150.0  150.0  250.0  250.0  350.0  …  1000.0  1150.0  1250.0  1350.0
```

A girdled tree of 10.32 m² takes a share of the larvae laid in its cell given by the ratio of the density of larvae on girdled to other trees. The article gives that ratio as `20.468 x^-0.559` at a density `x` per m², fitted at 16 outlier sites in Michigan; the code uses `18.25 (x + 1)^-0.533`, at least 1, which stays finite at low densities. The ratio falls as density rises: in a cell of 200 m², a girdled tree takes about half the eggs at the low densities ahead of a front, and 8% where the phloem is full.

```
x = 0:0.5:88
code = max.(18.25 .* (x .+ 1) .^ -0.533, 1)
text = 20.468 .* x .^ -0.559
taken(y) = 100 .* 10.32 .* y ./ (10.32 .* y .+ (200 - 10.32))
plot(plot(x, [code text]; ylabel="ratio, girdled to other trees", ylims=(0, 25),
          label=["code" "article"]),
     plot(x, [taken(code) taken(text)]; ylabel="share in the girdled tree (%)",
          ylims=(0, 100), label=["code" "article"]);
     xlabel="larvae per m² of phloem", lw=2, layout=(1, 2), size=(850, 320),
     left_margin=5Plots.mm, bottom_margin=5Plots.mm)
```

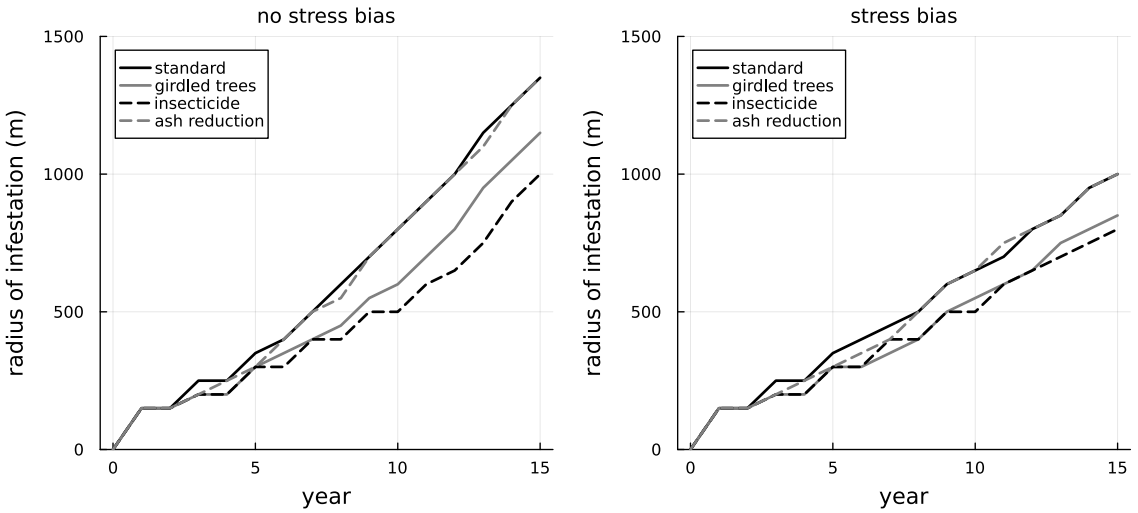


Under the stress bias, dispersing females favour cells with girdled trees (seven times as attractive, in every treated cell) or stressed trees, whose attraction is $7 \times 3 \times$ the share of phloem lost, kept between 1 and 7, so that a cell is fully attractive once it has lost a third of its phloem.

3 Treating within 300 m

The source's main comparison treats every cell within 300 m of the origin from the start and follows the infestation for 15 years.

```
treatments = (:none, :girdling, :insecticide, :ash_reduction)
labels = ["standard" "girdled trees" "insecticide" "ash reduction"]
runs = Dict((tr, sb) ⇒ simulate(; treatment=tr, stress_bias=sb)
            for tr in treatments for sb in (false, true))
plot([plot(0:15, [radii(runs[(tr, sb)]) for tr in treatments]; label=labels,
           title=sb ? "stress bias" : "no stress bias", titlefontsize=10,
xlabel="year",
           ylabel="radius of infestation (m)", lw=2, legend=:topleft, ylims=(0,
1500),
           color=[:black :gray :black :gray], ls=[:solid :solid :dash :dash])
      for sb in (false, true)]...; layout=(1, 2), size=(850, 380),
left_margin=5Plots.mm,
     bottom_margin=5Plots.mm)
```




```
code = Dict((:none, false) => (1350, 5.50), (:girdling, false) => (1150, 4.32),
            (:insecticide, false) => (1000, 3.64), (:ash_reduction, false) => (1350,
5.40),
            (:none, true) => (1000, 2.42), (:girdling, true) => (850, 2.11),
            (:insecticide, true) => (800, 1.82), (:ash_reduction, true) => (1000,
2.51))
article = Dict((:none, false) => (1350, 5.45), (:girdling, false) => (1150, 4.45),
               (:insecticide, false) => (950, 3.3), (:ash_reduction, false) =>
(1350, 5.3),
               (:none, true) => (1100, 2.5), (:girdling, true) => (950, 2.1),
               (:insecticide, true) => (850, 1.8), (:ash_reduction, true) => (1150,
2.8))
[(; treatment=tr, stress_bias=sb, radius=Int(radii(runs[(tr, sb)])[end]),
  code_radius=code[(tr, sb)][1], article_radius=article[(tr, sb)][1],
  beyond=round(eab_beyond(runs[(tr, sb)].u[end], 300) / 1e6; digits=2),
  code_beyond=code[(tr, sb)][2], article_beyond=article[(tr, sb)][2])
 for sb in (false, true) for tr in treatments]
```

```
8-element Vector{@NamedTuple{treatment::Symbol, stress_bias::Bool, radius::Int64,
code_radius::Int64, article_radius::Int64, beyond::Float64, code_beyond::Float64,
article_beyond::Float64}}:
 (treatment = :none, stress_bias = 0, radius = 1350, code_radius = 1350,
 article_radius = 1350, beyond = 5.48, code_beyond = 5.5, article_beyond = 5.45)
 (treatment = :girdling, stress_bias = 0, radius = 1150, code_radius = 1150,
 article_radius = 1150, beyond = 4.39, code_beyond = 4.32, article_beyond = 4.45)
 (treatment = :insecticide, stress_bias = 0, radius = 1000, code_radius = 1000,
 article_radius = 950, beyond = 3.64, code_beyond = 3.64, article_beyond = 3.3)
 (treatment = :ash_reduction, stress_bias = 0, radius = 1350, code_radius = 1350,
 article_radius = 1350, beyond = 5.33, code_beyond = 5.4, article_beyond = 5.3)
 (treatment = :none, stress_bias = 1, radius = 1000, code_radius = 1000,
 article_radius = 1100, beyond = 2.46, code_beyond = 2.42, article_beyond = 2.5)
 (treatment = :girdling, stress_bias = 1, radius = 850, code_radius = 850,
 article_radius = 950, beyond = 2.09, code_beyond = 2.11, article_beyond = 2.1)
 (treatment = :insecticide, stress_bias = 1, radius = 800, code_radius = 800,
 article_radius = 850, beyond = 1.79, code_beyond = 1.82, article_beyond = 1.8)
```

```
 (treatment = :ash_reduction, stress_bias = 1, radius = 1000, code_radius = 1000,
 article_radius = 1150, beyond = 2.51, code_beyond = 2.51, article_beyond = 2.8)
```

The radius after 15 years and the beetles beyond the treated area (millions), here, in the authors' code (run in R on the same settings) and in the article's Fig. 4, read by eye. The radii are the code's. As in the article:

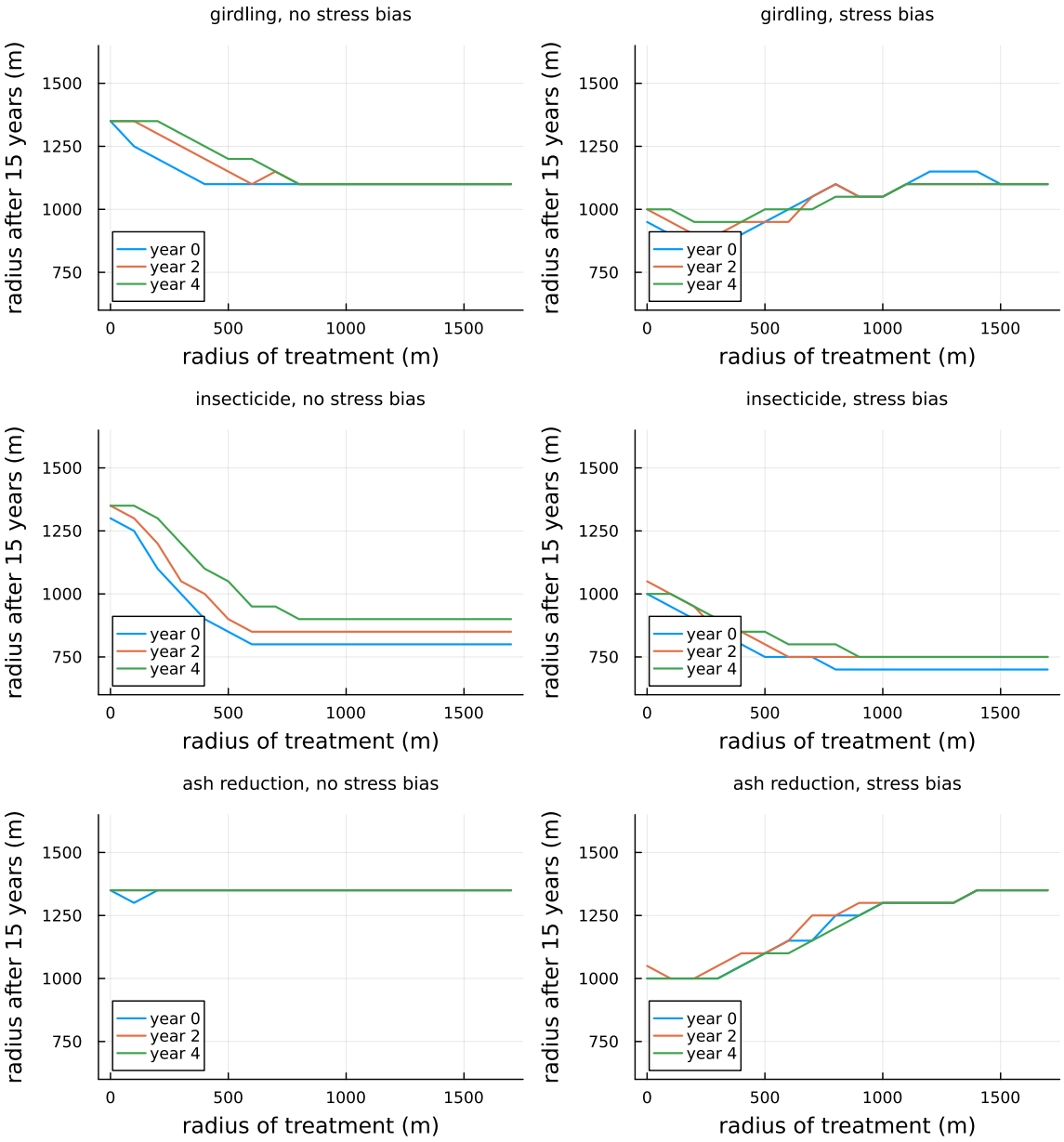
- insecticide slows the spread most, then girdling;
- ash reduction barely reduces the beetles beyond the treated area;
- the stress bias slows the spread, since dispersing females stay among the stressed trees near the origin.

The code itself departs from the article in two places: insecticide is a little weaker (1000 m against 950 without the bias), and with the stress bias every radius is 50–150 m smaller than the article's, although the beetles beyond the treated area agree except under ash reduction (2.5 million against 2.8).

4 The radius of treatment and the year it starts

The source then treated every cell out to radii of 0 to 1700 m, starting in the first year or up to four years later, when an outlier might be found.

```
rs = 0:100:1700
starts = (0, 2, 4)
cases = [(tr, sb, st, r) for tr in (:girdling, :insecticide, :ash_reduction)
         for sb in (false, true) for st in starts for r in rs]
final = Vector{Float64}(undef, length(cases))
Threads.@threads for i in eachindex(cases)
    tr, sb, st, r = cases[i]
    prob = problem(:eab_management; treatment=tr, stress_bias=sb, start=st,
radius=r)
    final[i] = eab_spread(solve(prob).u[end]).radius
end
sweep = Dict(cases .=> final)
plot([plot(rs, [[sweep[(tr, sb, st, r)] for r in rs] for st in starts];
           label=permutedims(["year $st" for st in starts]), lw=1.5,
           title="$(replace(String(tr), '_' => ' ')), $(sb ? "" : "no ")stress
bias",
           titlefontsize=9, ylims=(600, 1650), xlabel="radius of treatment (m)",
           ylabel="radius after 15 years (m)", legend=:bottomleft)
      for tr in (:girdling, :insecticide, :ash_reduction) for sb in (false,
true)]...;
     layout=(3, 2), size=(800, 860))
```



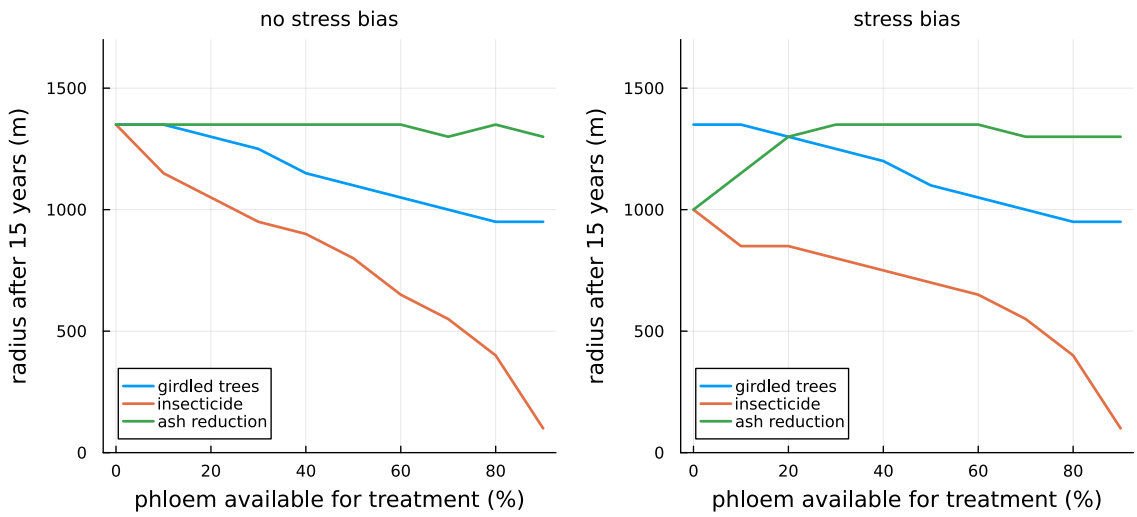
The results follow the source's Fig. 5:

- Insecticide: the radius falls to 800 m as the treated radius grows (700 m with the stress bias), as in the source. Later starts weaken it.
- Girdling: the radius falls to 1100 m once the treated circle passes 300–800 m, depending on the start, and stays there.
- Under the stress bias: girdling helps most at small radii (850 m at 200–300 m) and less at large ones, where the girdled cells draw beetles outwards, as the source found. Ash reduction, which never changes the radius without the bias, widens the spread with it as the treated circle grows, from 1000 m to 1350 m.

5 How many trees can be treated

The source varied the share of the phloem in large trees, and so the share that can be treated, from 0 to 90%. Here the whole lattice is treated.

```
shares = 0:0.1:0.9
share_cases = [(tr, sb, s) for tr in (:girdling, :insecticide, :ash_reduction)
               for sb in (false, true) for s in shares]
share_final = Vector{Float64}(undef, length(share_cases))
Threads.@threads for i in eachindex(share_cases)
    tr, sb, s = share_cases[i]
    prob = problem(:eab_management; treatment=tr, stress_bias=sb, share=s,
radius=1700)
    share_final[i] = eab_spread(solve(prob).u[end]).radius
end
by_share = Dict(share_cases .=> share_final)
plot([plot(100 .* shares,
           [[by_share[(tr, sb, s)] for s in shares]
            for tr in (:girdling, :insecticide, :ash_reduction)];
           label=["girdled trees" "insecticide" "ash reduction"], lw=2,
           title=sb ? "stress bias" : "no stress bias", titlefontsize=10,
           xlabel="phloem available for treatment (%)",
           ylabel="radius after 15 years (m)", ylims=(0, 1700), legend=:bottomleft)
      for sb in (false, true)]...; layout=(1, 2), size=(850, 380),
left_margin=5Plots.mm,
     bottom_margin=5Plots.mm)
```

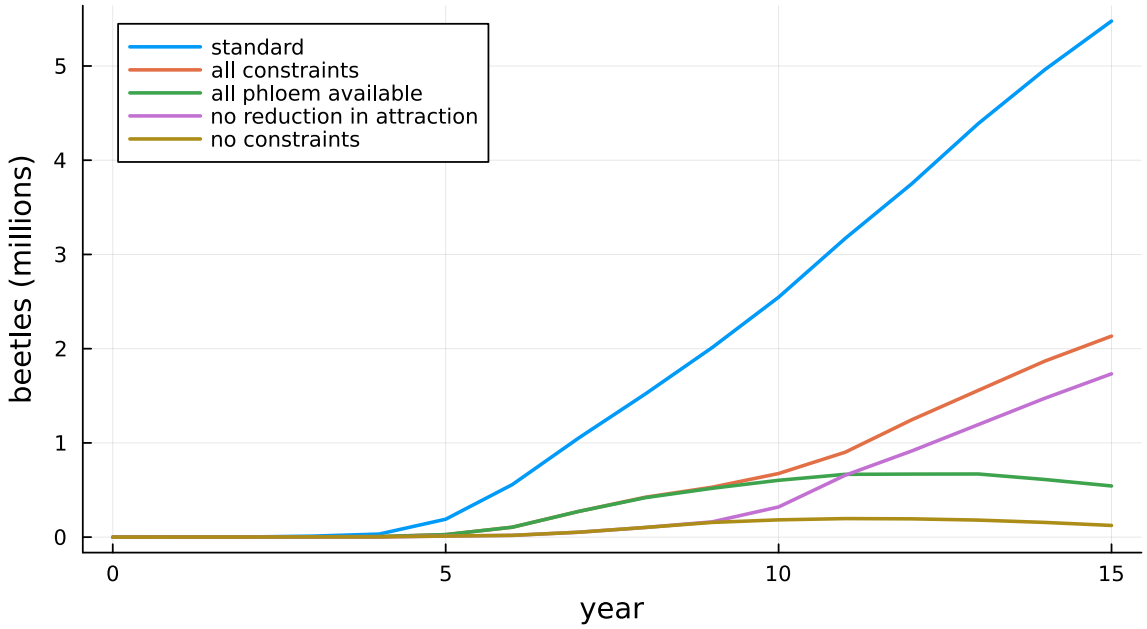


Insecticide's effect grows steadily with the share treated, to 100 m at 90% (about 200 m in the source's Fig. 3). Girdling helps less, because each cell girdles only one tree a year and the large trees run out. Ash removal barely changes the spread (1300–1350 m), as in the source.

6 The limits on girdling

Girdling is limited in two ways: only large trees can be girdled, and girdled trees draw fewer eggs as densities rise. The source's Fig. 6 lifts each limit in turn, girdling throughout the lattice.

```
scenarios = [("standard", (; treatment=:none)),
             ("all constraints", (; treatment=:girdling)),
             ("all phloem available", (; treatment=:girdling, share=1.0)),
             ("no reduction in attraction",
              (; treatment=:girdling, pmap=(; attraction_decline=false))),
             ("no constraints",
              (; treatment=:girdling, share=1.0, pmap=(;
attraction_decline=false)))]
beetles = [[eab_spread(u).beetles / 1e6 for u in simulate(; radius=1700, kw...).u]
           for (_, kw) in scenarios]
plot(0:15, beetles; label=permutedims(first.(scenarios)), lw=2, xlabel="year",
     ylabel="beetles (millions)", legend=:topleft, size=(650, 360))
```



Each limit lifted cuts the population further. With neither, the infestation stays small, as in the source.

7 The article's ratio

With the ratio printed in the article, which grows without bound at low densities, girdled trees take nearly all the larvae laid ahead of the front, and girdling within 300 m holds the radius to 1050 m rather than the code's 1150 m; capped at its value at 1 larva per m^2, it gives 1100 m.

```
article_ratio(cap) = (; girdle_a=20.468, girdle_shift=0.0, girdle_b=-0.559,
girdle_min=0.0,
                       girdle_max=cap)
[(; ratio=name,
  radius=eab_spread(solve(problem(:eab_management; treatment=:girdling,
pmap)).u[end]).radius)
 for (name, pmap) in (("code", (;)), ("article", article_ratio(Inf)),
                      ("article, capped", article_ratio(20.468)))]
```

```
3-element Vector{@NamedTuple{ratio::String, radius::Float64}}:
 (ratio = "code", radius = 1150.0)
 (ratio = "article", radius = 1050.0)
 (ratio = "article, capped", radius = 1100.0)
```

8 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:eab_management).deviations)
```

```
- The model follows the source's code, which departs from the article's text in
several ways, all kept: the ratio of larval density on girdled to other trees is
18.25 (x + 1)^-0.533, at least 1, where the text gives 20.468 x^-0.559 (girdle_a,
girdle_shift, girdle_b and girdle_max select the text's form, which needs a cap at
low densities); dispersal spans 40 cells rather than the base model's 50, and the
beetles of a cell holding no more than 0.9 are lost; the landscape has no border, so
dispersal is normalised within it.
- Under the stress bias, as in the code, a cell's attraction is 7 x lost / (max(A0,
3) / 3), kept within 1 and 7, where lost is the phloem lost since the start by any
means (the code's attempt to exclude removed phloem reads a year not yet computed
and has no effect), rather than rising linearly from 1 as the text suggests; every
treated cell under girdling attracts 7-fold from the first year, before treatment
starts and after its large trees are gone.
- As in the code, under ash reduction the adults landing on the large trees of a
treated cell are removed in the year treatment starts, the larvae are laid on all
the phloem, and the large trees left are removed after the larvae have fed, while
the larvae laid in them are kept; the code then sets every cell's large-tree phloem
to its whole phloem, which affects nothing else in its runs.
- As in the code, insecticide kills the larvae laid in a treated cell's large-tree
phloem (their density times its area), and feeding is then reduced again by the
unprotected share of the phloem.
- Large trees hold a share of each cell's phloem, as in the code, and a cell has
none where that share is at most 6.32 m²; a cell girdles one tree of 10.32 m² a year
while its large-tree phloem lasts.
- As in the base model, transposed = true reproduces the code's indexing of each
cell's beetles (from the cell with row and column swapped; nothing changes on the
symmetric landscapes of the source), and bare sets the phloem at or below which a
cell counts as without phloem (0 by default, as the code, where round-off decides;
1e-9 makes every eaten cell bare).
```

```
- The extra runs of the source's Appendix S2 (dispersal rates other than 0.037, and
landscapes with phloem drawn from a normal distribution) are not included; the
parameter b and the phloem keyword allow them.
```

Mercader, Rodrigo J., Nathan W. Siegert, Andrew M. Liebhold, and Deborah G. McCullough. 2011a. "Influence of Foraging Behavior and Host Spatial Distribution on the Localized Spread of the Emerald Ash Borer, *Agrilus Planipennis*." *Population Ecology* 53 (2): 271–85. https://doi.org/10.1007/s10144-010-0233-6.

Mercader, Rodrigo J., Nathan W. Siegert, Andrew M. Liebhold, and Deborah G. McCullough. 2011b. "Simulating the Effectiveness of Three Potential Management Options to Slow the Spread of Emerald Ash Borer (*Agrilus Planipennis*) Populations in Localized Outlier Sites." *Canadian Journal of Forest Research* 41 (2): 254–64. https://doi.org/10.1139/X10-201.